

Joseph Shetaye

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EDUCATION

Stanford University

Stanford, CA

Bachelor of Science in Computer Science, GPA: 4.04/4.00

Sep 2022 – Jun 2026 (expected)

Master of Science in Computer Science

Sep 2025 – Jun 2027 (expected)

Relevant Coursework: Artificial Intelligence: Principles and Techniques, Programming Languages, Operating Systems Principles, Embedded Operating Systems, Digital Systems Architecture, Advanced Systems Laboratory, Parallel Computing, Advanced Data Structures, Computer and Network Security, Compilers

TECHNICAL SKILLS

Programming Languages: C, C++, Swift, Rust, Python, CUDA, Go, Java, JavaScript, HTML, CSS, Dart

Tooling and Platforms: Linux, MacOS, Git, CMake, ARM ISA, Bash, Verilog, VHDL, PyTorch, Docker, Kubernetes, Flutter

WORK EXPERIENCE

Apple

Cupertino, CA

CoreOS Software Engineering Intern on the Darwin Server Team

Jun 2025 – Sep 2025

- Write low-level user space runtime services across all Apple platforms (MacOS, iOS, etc.).
- Contribute to Private Cloud Compute.

Stanford Computer Science Department

Stanford, CA

[CS 140E](#) & [CS 240LX](#) Section Leader

Jan 2025 – Jun 2025

- Ran a two-course sequence of hands-on operating system implementation courses.
- Focused on assisting students' lab understanding and teaching practical systems knowledge.

Stanford Artificial Intelligence Laboratory

Stanford, CA

Undergraduate Research Assistant at [Hazy](#)

Oct 2024 – Jun 2025

- Developed high performance attention variants for the Tokasaurus inference server.
- Designed a DSL for chip design and used reinforcement learning to improve LLM proficiency with the DSL.

To-be-announced Defense Startup

San Carlos, CA

Avionics and Software Intern

Feb 2024 – Jun 2024

- Participated in the entire software development lifecycle for counter UAS solutions.
- Owned the development of production real-time (5ms) guidance, navigation, and control software.
- Developed ultra-low latency computer vision pipelines for edge devices.

Tibex Technology

Minneapolis, MN

Full Stack Software Engineer

Dec 2019 – Sep 2023

- Developed an evidence discovery API integration that sends over 3,000 discoverable files from prosecutors to defense attorneys per month, significantly improving prosecutor and defense attorney productivity.
- Developed an integration with Axon body cameras that has processed over 7,000 video and photo files since implementation in December of 2022.

PROJECTS

Thread Equivalence Checker (C, ARMv6)

- Developed a verification tool to prove the correctness of multithreaded programs.
- Wrote optimized bare metal C with no OS support.
- Made extensive use of ARM architecture features such as the debug unit and MMU.

Satellite Flight Code (C, ARMv7-M, ARMv8-M)

- Developed satellite flight code from scratch (no off-the-shelf RTOS or drivers) to control binary size.
- Wrote peripheral and LoRa drivers.
- Implemented robust over-the-air updates.

Omega CPU (VHDL, Xilinx Spartan FPGA)

- Built a 32-bit RISC (reduced instruction set computer) CPU roughly based on the MIPS architecture.
- Presented the CPU at the Minneapolis MakerFair.

LEADERSHIP AND PROFESSIONAL DEVELOPMENT

Stanford Student Space Initiative

Stanford, CA

Satellite Team Member, Software Sub Team Lead

Sep 2023 – Jun 2025

- Collaborated with other students to build and develop reliable control software for satellite systems.
- Planned and executed rewrite of 12k+ line Python codebase in C.